

Towards Reliable and Generalizable Anomaly Detection



Invited Speaker

Jicong Fan

The Chinese University of Hong Kong

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Biography

Jicong Fan is currently an Assistant Professor with the School of Data Science, The Chinese University of Hong Kong, Shenzhen, China. He received the Ph.D. degree in electronic engineering from City University of Hong Kong, Hong Kong SAR, in 2018. From 2018 to 2020, he was a Postdoctoral Associate with the School of Operations Research and Information Engineering, Cornell University, Ithaca, NY, USA. He serves as an Associate Editor of Pattern Recognition and Neural Processing Letters, an Action Editor of Transactions on Machine Learning Research, and an Area Chair for ICML, ICLR, and NeurIPS. His research area is machine learning, with a particular focus on matrix and tensor methods, clustering, anomaly detection, and graph learning.

Abstract

Anomaly detection aims to identify unusual patterns with little or no anomaly supervision, making it challenging to define reliable detection criteria and generalize across real-world scenarios. In this talk, I will present our recent work on reliable and generalizable anomaly detection, including learning pseudo-anomalies, handling missing values and inherent noise, semi-supervised and fairness-aware detection, automated model selection, and universal outlier detection across heterogeneous datasets. I will also discuss extensions to graph and text anomaly detection. These studies aim to make anomaly detection more robust, practical, and broadly applicable.